

Tutorial - 1

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• Section-A (2x5=10)

1. What is a "Randomized algorithms"?

Ans Randomized algorithm in data structure and algorithms are algorithms that are used to randomize their computations to achieve a desired outcome. These algorithms introduce randomness to improve efficiency or simplify the algorithm design.

2. Find out GCD of 96 and 36.

Ans	Quotient	A	B	Reminders
	2	96	36	24
	1	36	24	12
	2	24	12	0
	x	12	0	x

GCD of 96 and 36 is : 12

3. Define approximate algorithm?

Ans An approximation algorithm is a way of dealing with NP-completeness for an optimization problem. This technique does not guarantee the best solution. The goal of approximation algo is to come as close as possible to the optimal solution in polynomial time.

4. Write Quotient Remainder theorem.

Ans The Quotient Remainder theorem states that for any integer a and any positive integer b , there exist unique integer q (quotient) and remainder (r) such that:

$$a = b \times q + r$$

$$\text{Where } 0 \leq r < b$$

5. Define Modular Multiplication.

Ans Modular Multiplication is a specific operation within modular arithmetic. Given two integers a and b , and a modulus n , the modular multiplication of a and b is defined as

$$(a \times b) \bmod n.$$

Section-3 ($3 \times 9 = 21$)

1. Explain Las Vegas algorithm with example.

Ans Las Vegas Algorithm:- A randomized algorithm which always produce a correct result or it informs about the failure is known as Las-Vegas Algo.

Example - Randomized Quick-sort.

Randomized Quick sort works by selecting a random element as the pivot element instead of choosing a fixed pivot.

This randomization aims to reduce the likelihood of worst case performance (which occurs when the pivot always end up being the smallest and largest elements).

2. Explain Monte Carlo Algorithm.

Ans Monte Carlo Algorithm:- The computational algorithms which rely on repeated random sampling to compute their results such algorithms are called a monte carlo algorithms. It can give the wrong answers, sometimes.

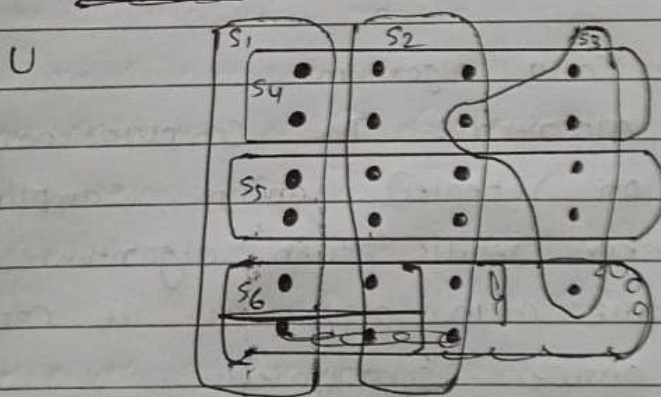
Example:- Estimating π using monte carlo
it is the method for estimating the value of π using random sampling. The idea is to randomly sample points in a square and determine how many fall within a quarter circle inscribed within the square.

Total no. of points

Ans Lower bound theory concept is based on the calculation of minimum is known as (lower bound) theory or Base Bound theory. The main aim to calculate a minimum number of comparisons require to execute an algorithm. Techniques which are used by lower bound Theory are :

- i) Comparison trees
- ii) Oracle and Adversary argument
- iii) State space method

1. Find minimum set cover.

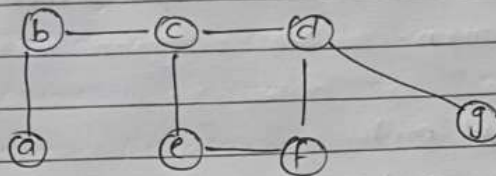


In first iteration set S_2 covers most elements that is 10 and hence it is added to set cover. In second iteration, set S_3 covers most elements i.e. 6 and 5 new elements

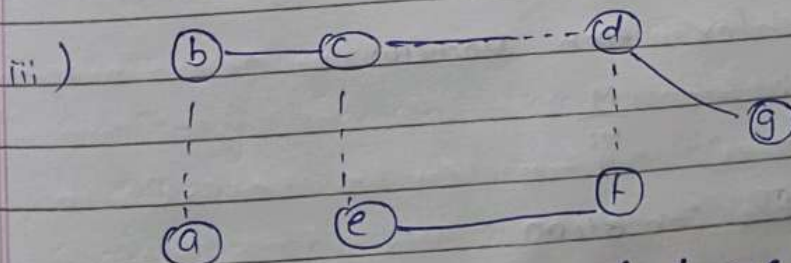
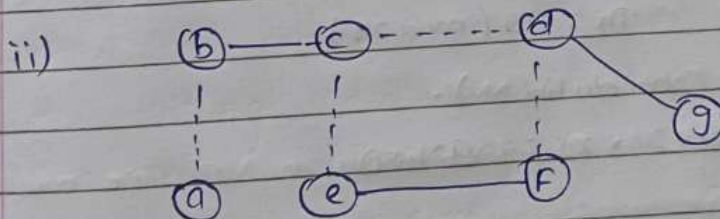
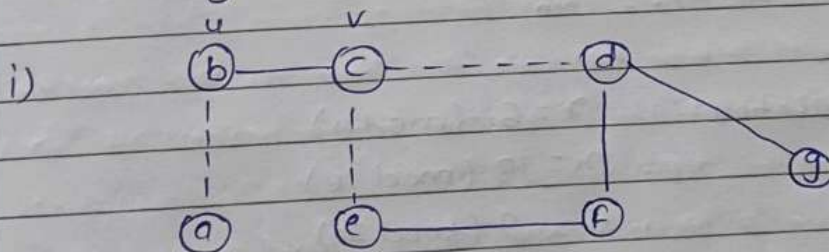
shot by Ankit Nov 30, 2024, 19:07 S3 added to set cover.

In third iteration Set S_1 covers all the remaining elements hence we add Set S_1 to set cover. The minimum number of sets that need to be selected to cover all the elements will be the optimal output i.e., Set cover = $\{S_2, S_3, S_1\}$.

2. Find vertex cover from below diagram.



Vertex cover: The idea is to take an edge (u,v) one by one, put both vertices to C and remove all the edges incident to u and v . We carry on until all edges have been removed.



vertex cover = $\{b, c, e, f, d, g\}$

1/20

State Chinese remainder theorem. Solve the following congruence.

$$x \equiv 6 \pmod{11}$$

$$x \equiv 13 \pmod{16}$$

$$x \equiv 9 \pmod{21}$$

$$x \equiv 19 \pmod{25}$$

Ans Chinese Remainder theorem (CRT):- it is a mathematical principle that solves systems of modular equations by finding a unique solution from the remainder of the division.

$$x = a_1 \pmod{m_1}$$

$$x = a_2 \pmod{m_2}$$

$$\vdots$$

$$x = a_n \pmod{m_n}$$

$$\text{then } x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_n M_n M_n^{-1}) \pmod{M}$$

$$\therefore M = M_1 \cdot M_2 \cdot \dots \cdot M_n$$

Given equations: $x \equiv 6 \pmod{11}$

$$x \equiv 13 \pmod{16}$$

$$x \equiv 9 \pmod{21}$$

$$x \equiv 19 \pmod{25}$$

Product of all moduli (M)

$$M = 11 \times 16 \times 21 \times 25 = 92400$$

Calculate M_i for each modulus

$$\text{i.e., } M_i = \frac{M}{m_i}$$

$$M_1 = \frac{92400}{11} = 8400$$

$$M_2 = \frac{92400}{16} = 5775$$

$$M_3 = \frac{92400}{21} = 4400$$

$$M_4 = \frac{92400}{25} = 3696$$

$$\text{Now, } M_1 \times M_1^{-1} = 1 \pmod{M_1}$$

$$8400 \times M_1^{-1} = 1 \pmod{11}$$

$$\text{Thus } M_1^{-1} = 8$$

$$\text{Similarly } M_2 \times M_2^{-1} = 1 \pmod{M_2}$$

$$5775 \times M_2^{-1} = 1 \pmod{16}$$

$$M_2^{-1} = 15$$

$$M_3 \times M_3^{-1} = 1 \pmod{21}$$

$$M_3^{-1} = 2$$

$$M_4 \times M_4^{-1} = 1 \pmod{25}$$

$$M_4^{-1} = 6$$

$$X = (6 \times 8400 \times 8 + 13 \times 5775 \times 15 + 9 \times 4400 \times 2 + 19 \times 3696 \times 6) \pmod{92400}$$

$$X = (38400 + 1126125 + 79200 + 421344) \pmod{92400}$$

$$X = (1665069) \pmod{92400}$$

$$X = 89469$$

Ans